

Post-Quantum Blind Signatures via Hash-Based Constructions:

A Systematic Survey and Gap Analysis

Literature Review — SPHINCS+ Poseidon2 Project

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Abstract

The transition to post-quantum cryptography (PQC) has produced standardised signature schemes — CRYSTALS-Dilithium, Falcon, and SPHINCS+ (SLH-DSA) — yet no post-quantum *blind signature* scheme has been standardised. Lattice-based blind signatures dominate the literature, achieving round-optimality via the Fischlin framework at flagship venues (CRYPTO, EUROCRYPT, CCS). Meanwhile, a 2026 generic compiler proves that any hash-and-sign signature can be converted into a round-optimal blind signature, but no concrete instantiation has been implemented or benchmarked. This survey systematically examines the four intersecting threads required for a hash-based blind signature: (i) SPHINCS+ as the underlying signature; (ii) the Fischlin generic construction; (iii) arithmetisation-oriented hash functions (Poseidon2, Monolith) for zero-knowledge circuit efficiency; and (iv) STARK proofs for transparent post-quantum attestation. We identify three critical voids in the literature: the absence of a concrete SPHINCS+-Fischlin instantiation, the absence of a Poseidon2-SPHINCS+ co-design analysis, and the absence of benchmarks comparing hash-based to lattice-based blind signatures in the Fischlin framework. A comparative analysis of 26 landmark papers across five dimensions (theory, implementation, security model, proof system, field arithmetic) reveals that the co-design of a single arithmetisation-oriented hash — Poseidon2 over the Goldilocks field — as both the SPHINCS+ tweakable hash and the STARK AIR substrate could eliminate the circuit-translation bottleneck that has blocked prior hash-based constructions.

Keywords: post-quantum cryptography, blind signatures, SPHINCS+, Fischlin framework, Poseidon2, STARK, zero-knowledge proofs, arithmetisation-oriented hash

1 Introduction

The NIST Post-Quantum Cryptography Standardisation process has selected three signature families for standardisation: CRYSTALS-Dilithium (lattice-based), Falcon (lattice-based), and SPHINCS+ (hash-based) Bernstein et al., 2019; NIST et al., 2022. These schemes guarantee *unforgeability* against quantum adversaries, yet standard digital signatures are inherently linkable: a verifier can associate every signature with its public key. This property is incompatible with applications requiring *unlinkability*, including anonymous credentials, e-voting schemes, privacy-preserving authentication, and off-chain attestation in distributed ledgers Buser et al., 2022; Chathurangi et al., 2025.

Blind signatures, introduced by Chaum (1983), allow a signer to produce a valid signature on a message without learning the message content. The user blinds the message before transmission; after receiving the blinded signature, the user unblinds it to obtain a standard signature that is unlinkable to the signing session. Blind signatures are the cryptographic backbone of privacy-preserving credentials and electronic cash.

The Fischlin framework Fuchsbaauer et al., 2015 provides a generic construction that compiles any suitable signature scheme into a round-optimal blind signature in the random-oracle model (ROM). The framework has been instantiated with lattice assumptions in a sequence of works del Pino and Katsumata, 2022; Beullens et al., 2023; Agrawal et al., 2022; Katsumata et al., 2021, producing the most efficient post-quantum blind signatures to date. Crucially, however, *all* these instantiations rely on structured lattice assumptions (Module-SIS, Module-LWE). A recent impossibility result Dietz et al., 2026 suggests fundamental barriers to pairing-free blind signatures in the ROM, further motivating the exploration of alternative assumptions.

A 2026 result by Bouillaguet et al. Bouillaguet et al., 2026 changes this landscape: they provide a generic compiler that transforms *any* post-quantum hash-and-sign signature into a blind signature with provable security in the ROM. The compiler applies to SPHINCS+, XMSS, and any hash-based signature, yet the authors do *not* provide a concrete instantiation or benchmarks. Herranz and Louiso Herranz and Louiso, 2025 independently explored the theoretical feasibility of hash-based blind signatures but likewise produced no implementation.

This survey. We systematically map the intersection of four research threads that must converge for a hash-based Fischlin blind signature:

1. **Hash-based signatures** — SPHINCS+, its WOTS+/FORS/hypertree architecture, and the properties required for the Fischlin compilation;
2. **The Fischlin framework** — generic compilation, security model, and the lattice-based instantiations that define the state of the art;
3. **Arithmetisation-oriented hash functions** — Poseidon2, Monolith, and competitors, evaluated for ZK-circuit efficiency and multi-instance security;
4. **STARK-based zero-knowledge proofs** — transparent post-quantum proof systems capable of attesting signature possession without trusted setup.

We analyse 26 landmark papers across these threads and identify three critical gaps that a convergent construction — SPHINCS+ instantiated with Poseidon2, compiled via the Bouillaguet Fischlin transform, and attested via Winterfell STARKs — would fill.

Roadmap. Section 2 reviews the three foundational primitives. Section 3 surveys the state of the art in post-quantum blind signatures, both lattice-based and hash-based. Section 4 catalogues arithmetisation-oriented hash functions and their ZK-circuit efficiency. Section 5 analyses STARK-based proof systems for signature attestation. Section 6 synthesises research gaps and positions the SPHINCS+–Poseidon2–Fischlin construction. Section 7 provides a structured comparative analysis across five dimensions. Section 8 concludes with recommendations for future work.

2 Foundational Primitives

2.1 SPHINCS+: Stateless Hash-Based Signatures

SPHINCS+ Bernstein et al., 2019, standardised as NIST FIPS 205 (SLH-DSA), is a *stateless* hash-based signature scheme. Unlike stateful schemes (XMSS, LMS) addressed in NIST SP 800-208 NIST et al., 2020, SPHINCS+ requires no state tracking between signing operations, making it suitable for deployment scenarios where state synchronisation is impractical.

The scheme combines three layers of hash-based constructs:

- **WOTS+** (Winternitz One-Time Signature Plus): a one-time signature whose security rests solely on second-preimage resistance of the underlying hash function;
- **FORS** (Forest of Random Subsets): a few-time signature that replaces HORST, offering tighter security bounds for the same hash budget;
- **XMSS-MT** hypertree: a multi-tree Merkle structure that compresses many WOTS+ public keys into a single root, which serves as the SPHINCS+ public key.

SPHINCS+ signs a message by selecting a random leaf in the hypertree and authenticating the path through FORS and WOTS+ verification. Every operation is purely hash-based, with no number-theoretic assumptions. The main drawback is signature size (typically 7–50 KB depending on security level), though NIST IR 8413 NIST et al., 2022 concludes this is acceptable for many applications.

For blind-signature purposes, the critical property is that SPHINCS+ is a *hash-and-sign* scheme: the signer hashes the message via a tweakable hash (THASH) and then signs the hash with the hypertree. This structure is directly amenable to the Fischlin compilation Boullaguet et al., 2026.

2.2 The Fischlin Blind-Signature Framework

Fischlin (2006) proposed a generic construction transforming any three-move identification scheme into a round-optimal (two-move) blind signature. Fuchsbauer et al. Fuchsbauer et al., 2015 refined this to standard-model security under the DLIN assumption; subsequent works extended it to the lattice setting del Pino and Katsumata, 2022; K stner et al., 2024.

The modern Fischlin construction for a hash-and-sign signature Σ proceeds as follows:

Phase 1. Commit. The user samples randomness r , blinds message m as $\bar{m} = H(m, r)$, and sends $\bar{m}$ to the signer.

Phase 2. Issue. The signer computes $\sigma_{\text{blind}} \leftarrow \Sigma.\text{Sign}(sk, \bar{m})$ and returns it.

Phase 3. Finalise credential. The user unblinds σ_{blind} to obtain σ on m , and stores the witness (m, r, σ) .

Phase 4. Show. The user generates a zero-knowledge proof π that they know a valid opening (m, r) such that $\Sigma.\text{Verify}(pk, m, \sigma) = 1$ and $\bar{m} = H(m, r)$ is the message the signer actually signed.

Security requires *one-more unforgeability* (OMUF) — the user cannot produce more valid signatures than signing sessions — and *blindness* — the signer cannot link a finalised signature to a particular session — both in the ROM.

2.3 Arithmetisation-Oriented Hash Functions

Standard hash functions (SHA-256, SHAKE-256) require thousands of algebraic constraints per invocation in a ZK circuit, making a full SPHINCS+ verification proof impractical. Arithmetisation-oriented (AO) hash functions solve this by operating natively over large prime fields, reducing the constraint cost by orders of magnitude.

Poseidon2

Poseidon2 Grassi et al., 2023, the successor to Poseidon Grassi et al., 2019, is a family of AO hash functions designed for optimal efficiency in ZK proof systems. The canonical instantiation operates as a sponge construction over the Goldilocks prime field $\mathbb{F}_p$ with $p = 2^{64} - 2^{32} + 1$, a field chosen for its efficient modular reduction and native support in STARK provers such as Winterfell.

The Poseidon2 permutation has width $t = 12$ (rate $r = 6$, capacity $c = 6$), with $R_F = 8$ full rounds and $R_P = 22$ partial rounds. The non-linear layer is $x \mapsto x^3$, which is a permutation over $\mathbb{F}_p$ when $\gcd(3, p-1) = 1$. Each Poseidon2 permutation requires approximately 300 R1CS constraints, compared to approximately 25,000 for a SHA-256 compression Grassi et al., 2024.

Competitors

Monolith Grassi et al., 2024 achieves native CPU performance comparable to SHA3-256 while maintaining ZK-friendliness, making it superior for CPU-bound verification but slightly less efficient than Poseidon2 for pure constraint-minimisation use cases. Poseidon(2)b Grassi et al., 2026 extends the design to binary extension fields ($\mathbb{F}_{2^k}$), targeting proving systems such as Binius. Kovalchuk et al. Kovalchuk et al., 2021 analyse Poseidon against non-binary differential and linear attacks, confirming that the wide-trail strategy provides the expected security margins when parameters are chosen appropriately.

For the specific use case of SPHINCS+ verification in a STARK circuit, where all operations are over the same prime field, Poseidon2’s field-native design is optimal.

2.4 STARK Proofs

STARKs (Scalable Transparent ARguments of Knowledge) Ben-Sasson et al., 2020 are zero-knowledge proof systems with three distinguishing properties:

- **Transparent:** no trusted setup; security relies solely on collision-resistant hash functions;
- **Post-quantum:** symmetric-key assumptions are quantum-resistant by design;
- **Scalable:** proof size and verification time grow poly-logarithmically with computation size.

Table 1: Comparison of Post-Quantum Blind Signature Schemes

Scheme	Rounds	Assumption
Fuchsbauer et al. (2015) Fuchsbauer et al., 2015	2	DLIN
del Pino–Katsumata (2022) del Pino and Katsumata, 2022	2	Module-SIS/LWE
Agrawal et al. (2022) Agrawal et al., 2022	2	Module-SIS/LWE
Beullens et al. (2023) Beullens et al., 2023	2	Ring/Module-SIS/LWE + NTRU
Kastner et al. (2024) Kästner et al., 2024	2	SIS (pairing-free)
Katsumata et al. (2021) Katsumata et al., 2021	2	LWE + ABET
Bouillaguet et al. (2026) Bouillaguet et al., 2026	2	Any hash-and-sign
<i>Target (this project)</i>	<i>2</i>	<i>Hash (SPHINCS+)</i>

Note. “Size” refers to the final unblinded signature size (or proof-of-possession size for Fischlin-style schemes). ROM = random oracle model. ABET = attribute-based encryption with threshold. Benchmarked sizes are approximate and parameter-dependent.

The STARK protocol proceeds in three phases: (1) *Arithmetisation* — the computation is expressed as an algebraic execution trace and a set of polynomial constraints (AIR); (2) *Low-degree extension* — the trace is extended using Reed-Solomon encoding over a multiplicative subgroup of $\mathbb{F}_p$; (3) *FRI protocol* — a sequence of folding steps reduces the degree check to a constant number of Merkle-tree openings, achieving logarithmic proof size.

Bulletproofs Bünz et al., 2018 and MPC-in-the-head techniques Katz et al., 2018 offer alternative proof systems, but both rely on discrete-log assumptions (Bulletproofs) or produce proofs linear in circuit size (MPC-in-the-head), making STARKs the preferred choice for transparent post-quantum attestation.

3 Post-Quantum Blind Signatures: State of the Art

3.1 Lattice-Based Constructions

The dominant post-quantum approach builds blind signatures on lattice problems (SIS/LWE). Table 1 summarises the landmark constructions.

Several observations emerge from the lattice-based literature. First, *round-optimality is achievable*: all recent schemes achieve two-round interaction, confirming the Fischlin framework’s practical viability Fuchsbauer et al., 2015; del Pino and Katsumata, 2022. Second, *signature sizes are competitive*: Beullens et al. Beullens et al., 2023 achieve approximately 20 KB signatures, approaching the size of RSA blind signatures, albeit at the cost of an ad-hoc NTRU assumption. Third, *standard-model security is expensive*: Katsumata et al. Katsumata et al., 2021 avoid random oracles but produce approximately 100 KB signatures and require hierarchical inner-product functional encryption. Finally, *verifier-side computation is dominated by the zero-knowledge proof*: in the Fischlin show phase, the bottleneck is the ZK proof verification regardless of the underlying signature scheme.

3.2 Hash-Based Blind Signatures: The Emerging Frontier

Two recent papers directly address the possibility of hash-based blind signatures and collectively define a new research direction.

Herranz and Louiso Herranz and Louiso, 2025 provide the first systematic exploration of hash-based blind signatures. They analyse the feasibility of instantiating the Fischlin framework with XMSS and SPHINCS+, identifying three challenges: (i) the determinism of hash-based signing (SPHINCS+ signs each message with a fresh random leaf, which is beneficial for blindness); (ii) the large signature size affecting proof size; and (iii) the need for a ZK-friendly hash to make the STARK circuit efficient. Their work establishes the problem statement but stops short of a concrete instantiation.

Bouillaguet, Feneuil, Maire, Rivain, Sauvage, and Vergnaud Bouillaguet et al., 2026 present the most significant advance to date: a generic compiler that transforms *any* post-quantum hash-and-sign signature scheme into a blind signature under the Fischlin framework, provably secure in the ROM. Their construction achieves:

- **Blindness by design:** the commitment mechanism is statistically hiding;
- **One-more unforgeability:** reduces to the EUF-CMA security of the underlying signature scheme;
- **Round-optimality:** two-round protocol (optimal);
- **Applicability:** works with SPHINCS+, XMSS, and any hash-and-sign signature.

Crucially, the authors do *not* provide an implementation or benchmarks of the compiler instantiated with SPHINCS+, nor do they analyse the ZK-circuit cost of the resulting proof.

Blind signatures are the core building block of anonymous credential systems. Two surveys map this landscape comprehensively. Buser et al. Buser et al., 2022 provide a systematic survey of exotic post-quantum signatures (blind, ring, group, adaptor, threshold) for blockchain applications, identifying blind signatures as the most deployment-ready primitive for privacy-preserving interactions. Chathurangi et al. Chathurangi et al., 2025 trace the evolution of anonymous credentials from traditional (RSA, pairing-based) to post-quantum (lattice, code-based), confirming that *no hash-based anonymous credential system* has been proposed. Slamanig Slamanig, 2025 discusses the theory-practice gap in privacy-preserving authentication, noting that existing standards (ISO/IEC 29191, W3C Verifiable Credentials) rely on pairing-based constructions vulnerable to quantum attacks.

Argo et al. Argo et al., 2024 implement and benchmark post-quantum signatures (blind, group, ring) using lattice assumptions, achieving practical performance. Their work demonstrates that PQ blind signatures can be deployed with reasonable overhead, but it remains confined to the lattice setting.

3.3 Theoretical Barriers

Dietz et al. Dietz et al., 2026 prove an impossibility result for round-optimal pairing-free blind signatures in the standard model, raising questions about the feasibility of hash-based constructions. However, their result applies to the *standard model*, not the ROM, and does not rule out constructions that exploit hash-specific properties (e.g., the Fischlin compiler’s reliance on the ROM). Kästner et al. Kästner et al., 2024 construct a pairing-free scheme in the ROM, demonstrating the boundary of the impossibility result.

Table 2: Comparison of Arithmetisation-Oriented Hash Functions

Hash	Field	Constraints/perm	Security	CPU speed
Poseidon2 Grassi et al., 2023	$\mathbb{F}_p$ (Goldilocks)	~ 300 R1CS	128-bit	Moderate
Monolith Grassi et al., 2024	$\mathbb{F}_p$ (Goldilocks)	~ 400 R1CS	128-bit	Fast ($\approx$ SHA3)
Poseidon(2)b Grassi et al., 2026	$\mathbb{F}_{2^k}$	~ 250 R1CS	128-bit	Moderate
Poseidon Grassi et al., 2019	$\mathbb{F}_p$	~ 350 R1CS	128-bit	Moderate
SHA-256 (baseline)	Binary	$\sim 25,000$ R1CS	128-bit	Fast (ASIC)

Note. Constraint counts are approximate for a single permutation (Poseidon2, Monolith, Poseidon) or single compression (SHA-256). CPU speed is qualitative based on published benchmarks. Poseidon(2)b targets binary-field proving systems (Binius), not the Goldilocks field used in Winterfell.

4 Arithmetisation-Oriented Hash Functions and their ZK Efficiency

The efficiency of a STARK-based Fischlin proof is dominated by the hash function used inside the AIR circuit. Standard hash functions impose a prohibitive constraint cost: a single SHA-256 compression function requires approximately 25,000 R1CS constraints, making a full SPHINCS+ verification proof (approximately 2,000 hash calls) infeasible at approximately 50 million constraints. AO hash functions reduce this by operating natively over large prime fields.

4.1 Poseidon2: Optimal Constraint Efficiency

Poseidon2 Grassi et al., 2023 achieves the best constraint-to-security ratio among current AO hashes for the Goldilocks field. Table 2 compares the relevant metrics.

Monolith Grassi et al., 2024 outperforms Poseidon2 in native CPU speed (comparable to SHA3-256) but has slightly worse constraint efficiency for the Merkle-tree use case. For pure ZK-proving, where constraint count dominates proving time, Poseidon2 is preferred.

4.2 Co-Design Thesis

The central insight for hash-based Fischlin blind signatures is *co-design*: by using Poseidon2 as the tweakable hash (THASH) inside SPHINCS+, the same hash function serves dual purposes:

1. SPHINCS+ signature verification (FORS tree construction, WOTS+ chains, Merkle path authentication);
2. STARK AIR constraints (Poseidon2 permutation equality checks, sponge continuity, Merkle root assertions).

This eliminates the need for a *hash function translation layer* that would bloat the circuit by orders of magnitude. Adomnicai Adomnicai, 2026 explores a similar co-design for multi-party hash chain applications, confirming the feasibility of Poseidon2-based hash chains across multiple protocol layers.

Quan Quan, 2022 demonstrates a related approach combining lattice-based aggregate signatures with STARK protocols, though the setting uses two distinct hash functions and does not achieve the co-design efficiency of a unified AO substrate.

Table 3: Comparison of Zero-Knowledge Proof Systems for Signature Attestation

Proof System	Assumption	Proof Size	Trusted Setup	PQ-Secure
STARK (Winterfell)	Hash (Poseidon2)	~85 KB	None	Yes
Bulletproofs Bünz et al., 2018	Discrete Log	~1 KB	None	No
MPC-in-the-head Katz et al., 2018	Hash (SHA-256)	~1 MB	None	Yes
Groth16 (SNARK)	Pairing (BLS12-381)	~200 B	Required	No

Note. Proof sizes are indicative and parameter-dependent. STARK size assumes blowup factor 16 at 128-bit security. Bulletproofs and Groth16 rely on discrete-log and pairing assumptions respectively, both vulnerable to quantum attacks via Shor’s algorithm. MPC-in-the-head produces proofs linear in circuit size (~1 MB for SPHINCS+ verification).

5 STARK-Based Zero-Knowledge Proofs for Signature Attestation

5.1 Why STARKs Over Alternatives

The Fischlin *Show* phase requires the holder to prove, in zero-knowledge, possession of a valid signature. The proof system must satisfy four requirements:

- **Post-quantum secure:** cannot rely on discrete-log or pairing assumptions;
- **Transparent:** no trusted setup (the signer may be malicious);
- **Succinct:** proof size must be practical for bandwidth-constrained verification;
- **Field-native:** the AIR should operate over the same field as the hash function.

Table 3 compares the leading candidates.

STARKs are the only option satisfying all four requirements simultaneously. Bulletproofs Bünz et al., 2018 are succinct but rely on the discrete-log assumption, which Shor’s algorithm breaks. MPC-in-the-head techniques Katz et al., 2018 avoid trusted setup and can be post-quantum, but produce proofs linear in the circuit size — for SPHINCS+ verification at approximately 600,000 R1CS constraints, the resulting proof would be on the order of megabytes, impractical for bandwidth-constrained applications.

5.2 Winterfell Full-AIR Approach

Winterfell provides a Rust-based STARK prover with native support for the Goldilocks field. For SPHINCS+ verification, the full-AIR approach encodes the entire verification trace as an algebraic execution tableau. The AIR enforces:

1. Poseidon2 permutation constraints: rate-lane and capacity-lane equality across rounds, round-constant binding, and partial-round S-box constraints;
2. Sponge continuity: consecutive permutation outputs match inputs across the hash chain;
3. Message absorption: the SPHINCS+ message and public key are bound to the trace;
4. Root assertion: the computed Merkle root matches the signer’s public key;

5. Round counter integrity: prevents permutation skipping or reordering.

The total constraint count is 53 (boundary + transition) for a trace of 23,861 rows $\times$ 64 columns at 128-bit security, encoding 2,091 Poseidon2 permutations. Proof size is approximately 85 KB, with proving time of approximately 37 seconds on a laptop (blowup factor 16) and verification in approximately 4.2 ms Adomnicai, 2026.

5.3 Proof-System Overhead in the Fischlin Protocol

In a Fischlin blind-signature protocol, the ZK proof is the dominant cost of the *Show* phase. The signer-side operations (blind signature issuance) are unaffected by the choice of proof system. For the holder, the proving time determines usability: 37 seconds is acceptable for infrequent credential issuance (e.g., once per epoch) but prohibitive for high-frequency use. For the verifier, the 4.2 ms verification and 85 KB proof size are practical for most deployment scenarios, including blockchain-based verification where gas costs scale with proof size Quan, 2022.

6 Research Gaps and Project Positioning

The preceding sections reveal a consistent pattern: each of the four required threads has independent maturity, but their intersection remains unexplored. We identify three critical gaps.

6.1 G-1: Absence of a Concrete SPHINCS+–Fischlin Instantiation

The Bouillaguet compiler Bouillaguet et al., 2026 proves that any hash-and-sign scheme can be compiled into a blind signature. Herranz and Louiso Herranz and Louiso, 2025 establish theoretical feasibility. Yet no concrete implementation of a SPHINCS+–Fischlin blind signature exists as of mid-2026. The compiler’s security proof is in the ROM and the construction applies directly to SPHINCS+, but the deployability depends on engineering choices — witness encoding, trace layout, AIR constraint design — that the theoretical papers do not address.

Project response. This project implements the first SPHINCS+–Fischlin blind signature by instantiating the Bouillaguet compiler with SPHINCS+ over Poseidon2. The implementation includes a full Fischlin protocol stack (commit, issue, finalise, show, verify) with a Winterfell-based STARK prover for the Show phase.

6.2 G-2: Absence of Poseidon2–SPHINCS+ Co-Design Analysis

The original SPHINCS+ specification Bernstein et al., 2019 defines five hash backends (SHA-256, SHAKE-256, Haraka, and their variants) but does not include an arithmetisation-oriented hash. Replacing the SPHINCS+ hash backend with Poseidon2 requires:

1. Adapting the tweakable hash (THASH) abstraction to Poseidon2’s sponge construction;
2. Verifying that Poseidon2 satisfies the multi-instance security properties required for SPHINCS+’s security reduction (SM-TCR, SM-DSPR, SM-PRE, SM-UD) in the ROM;

3. Demonstrating that the constraint efficiency gain (approximately $80\times$ fewer constraints per hash call compared to SHA-256) translates to a feasible STARK proof.

No prior work systematically addresses all three sub-requirements.

Project response. This project provides the first complete co-design of Poseidon2 as the SPHINCS+ THASH backend, including the adapter layer (`hash_poseidon2_adapter`), the parameter selection for 128-bit and 192-bit security levels, and the STARK AIR constraints that operate over the same Poseidon2 permutation.

6.3 G-3: Absence of Hash-Based vs. Lattice Blind-Signature Benchmarks

The post-quantum blind-signature literature is almost exclusively lattice-based del Pino and Katsumata, 2022; Beullens et al., 2023; Agrawal et al., 2022. The natural comparison — hash-based vs. lattice-based for equivalent security levels, measuring signature size, proving time, verification time, and proof size — has not been performed. Without such benchmarks, protocol designers lack the data needed to choose between assumptions.

Project response. This project benchmarks the SPHINCS+–Poseidon2–Fischlin construction at 128-bit and 192-bit security levels and compares the results against published lattice benchmarks. Metrics include proof size, prover time, verifier time, and total protocol bandwidth.

6.4 Summary of the Three-Layer Architecture

The project’s theoretical architecture comprises three stacked layers, unified by a single arithmetisation-oriented hash:

1. **Layer 1: SPHINCS+** — provides EUF-CMA-secure hash-based signatures over Poseidon2, operating on the Goldilocks field $\mathbb{F}_p$;
2. **Layer 2: Fischlin Transform + STARK** — compiles SPHINCS+ into a round-optimal blind signature, using a STARK proof (via Winterfell) for the Show phase;
3. **Layer 3: Poseidon2 over Goldilocks** — serves as the unified algebraic substrate, acting simultaneously as the SPHINCS+ tweakable hash, the STARK AIR permutation, and the FRI low-degree test oracle.

The novelty lies in the *unification*: this is the first time a single AO hash serves as the unified substrate across all three layers of a blind-signature protocol, eliminating the hash-function translation bottleneck that has historically made hash-based Fischlin constructions prohibitively expensive.

7 Comparative Analysis across Five Dimensions

We evaluate the 26 surveyed papers across five analytical dimensions. Table 4 summarises the results.

- D1. Theory foundation.** Does the paper provide a security proof in a standard cryptographic model (ROM, standard model, QROM)?
- D2. Implementation.** Does the paper provide runnable code or benchmarks?
- D3. Security model.** What security properties are proven (EUF-CMA, OMUF, blindness, anonymity)?
- D4. Proof system.** If applicable, what ZK proof system is used (STARK, Bulletproofs, MPC-in-the-head, none)?
- D5. Field arithmetic.** What algebraic structure underlies the construction (lattice, pairing, hash, $\mathbb{F}_p$)?

7.1 Analysis of the Coverage Map

Table 4 reveals a clear cluster in the lattice region (D5): all six Fischlin blind-signature constructions at flagship venues (CRYPTO, CCS, EUROCRYPT) are lattice-based. The hash-based row — Bouillaguet et al. Bouillaguet et al., 2026 — provides the generic theory (D1) but lacks implementation and benchmarks (D2). Herranz and Louiso Herranz and Louiso, 2025 are the only paper focused exclusively on hash-based blind signatures, but their contribution is limited to feasibility analysis.

The AO hash cluster (Grassi et al. Grassi et al., 2023, 2019, 2024, 2026) provides implementation and benchmarks for the hash function itself but does not connect to the blind-signature or Fischlin literature. Adomnicai Adomnicai, 2026 bridges AO hashes to protocol-level constructions but focuses on multi-party computation rather than blind signatures.

The intersection marked by the project target row — hash-based signature $\cap$ Fischlin blind signature $\cap$ AO hash $\cap$ STARK proof — is empty in the existing literature.

8 Conclusion

This survey has mapped the four converging research threads necessary for a hash-based Fischlin blind signature: SPHINCS+ hash-based signatures, the Fischlin blind-signature framework, arithmetisation-oriented hash functions (Poseidon2), and STARK-based zero-knowledge proofs. Our analysis of 26 landmark papers across these threads yields five principal findings.

Finding 1: A generic compiler exists but is uninstantiated. Bouillaguet et al. Bouillaguet et al., 2026 (IEEE S&P 2026) provide a provably secure generic compiler from any hash-and-sign signature to a round-optimal blind signature in the ROM. The compiler applies directly to SPHINCS+, but no concrete instantiation or benchmark has been published.

Finding 2: Lattice dominance is complete but not inherent. Six flagship-venue constructions Fuchsbaauer et al., 2015; del Pino and Katsumata, 2022; Beullens et al., 2023; Agrawal et al., 2022; Katsumata et al., 2021; Kästner et al., 2024 all use lattice assumptions. The Bouillaguet compiler demonstrates that the Fischlin framework does *not* require lattices, but the hash-based pathway remains unexplored.

Finding 3: Poseidon2–SPHINCS+ co-design is optimal but unvalidated. Using Poseidon2 as both the SPHINCS+ THASH backend and the STARK AIR substrate eliminates the circuit-translation bottleneck, reducing constraint count by approximately $80\times$ compared to SHA-256. The multi-instance security properties required for SPHINCS+’s reduction (SM-TCR, SM-DSPR, SM-PRE, SM-UD) have not been formally verified for Poseidon2 in the ROM, though Grassi et al. Grassi et al., 2023, 2019 provide strong evidence they hold.

Finding 4: STARK proving costs are manageable for credential use cases. At approximately 37 seconds prover time and 85 KB proof size (128-bit security), the Winterfell full-AIR approach is practical for non-interactive use cases such as credential issuance verified once per epoch. The verification time of approximately 4.2 ms supports high-throughput verifier deployments.

Finding 5: Hash-based blind signatures occupy a unique and valuable niche. They are the only post-quantum blind-signature approach relying solely on hash-function security, without structured lattice assumptions, thereby offering the most conservative security profile for high-assurance privacy applications. Chathurangi et al. Chathurangi et al., 2025 and Slamanig Slamanig, 2025 confirm that no hash-based anonymous credential system currently exists, underscoring the novelty of the construction.

8.1 Recommendations for Future Work

Based on the gaps identified, we recommend the following research priorities:

1. **Implement and benchmark the SPHINCS+–Poseidon2–Fischlin construction.** The Bouillaguet compiler Bouillaguet et al., 2026 provides the theoretical foundation; what is needed is a concrete implementation measuring proof size, prover time, verifier time, and total protocol bandwidth at standardised security levels.
2. **Formally verify Poseidon2’s THF properties for SPHINCS+.** The four properties required by SPHINCS+’s security reduction (SM-TCR, SM-DSPR, SM-PRE, SM-UD) should be formally verified for Poseidon2 in the ROM, following the modular THF framework used in the SPHINCS+ specification.
3. **Optimise STARK proving time.** At approximately 37 seconds, the prover is the bottleneck. Recursive proof composition, hardware acceleration of Poseidon2 permutations, and parameter optimisation (fewer FORS trees, shorter hypertrees) are promising directions.
4. **Establish post-quantum blind-signature standards.** The IETF CFRG and NIST should include advanced primitives — blind signatures, group signatures, anonymous credentials — in the PQC standardisation roadmap. A hash-based option would provide the most conservative security foundation for these standards.

Closing remark. The convergence of SPHINCS+, Poseidon2, the Fischlin framework, and STARK proofs represents a unique opportunity to construct the first practical hash-based blind-signature scheme — a primitive that combines post-quantum security, privacy, transparency, and

the conservative assurance of hash-function-only assumptions. The gaps identified in this survey are addressable with current technology; filling them would deliver a blind-signature scheme with no trusted setup, no lattice assumptions, and native STARK compatibility.

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Table 4: Five-Dimension Analysis of the 26-Paper Corpus

Paper	Core contribution	D1	D2	D3
Bernstein et al. (2019) Bernstein et al., 2019	SPHINCS+ signature framework	Proof	Impl	EUUF-CMA
NIST IR 8413 (2022) NIST et al., 2022	Third-round PQC status	Report	—	—
NIST SP 800-208 (2020) NIST et al., 2020	Stateful HBS standard	Report	—	EUUF-CMA
Fuchsbaauer et al. (2015) Fuchsbaauer et al., 2015	Round-optimal blind sigs	Proof	—	OMUF+BL
del Pino–Katsumata (2022) del Pino and Katsumata, 2022	Lattice Fischlin framework	Proof	—	OMUF+BL
Beullens et al. (2023) Beullens et al., 2023	Short lattice blind sigs	Proof	—	OMUF+BL
Agrawal et al. (2022) Agrawal et al., 2022	Practical lattice blind sigs	Proof	Bench	OMUF+BL
Katsumata et al. (2021) Katsumata et al., 2021	Standard-model blind sigs	Proof	—	OMUF+BL
Kastner et al. (2024) Kästner et al., 2024	Pairing-free blind sigs	Proof	—	OMUF+BL
Dietz et al. (2026) Dietz et al., 2026	Impossibility result	Proof	—	—
Bouillaguet et al. (2026) Bouillaguet et al., 2026	Generic Fischlin compiler	Proof	—	OMUF+BL
Herranz–Louiso (2025) Herranz and Louiso, 2025	Hash-based BS feasibility	Theory	—	—
Grassi et al. (2023) Grassi et al., 2023	Poseidon2 design	Proof	Impl	—
Grassi et al. (2019) Grassi et al., 2019	Poseidon design	Proof	Impl	—
Grassi et al. (2024) Grassi et al., 2024	Monolith design	Proof	Impl	—
Grassi et al. (2026) Grassi et al., 2026	Poseidon(2)b design	Proof	Impl	—
Adomnicai (2026) Adomnicai, 2026	AO hash chains	Proof	Impl	—
Kovalchuk et al. (2021) Kovalchuk et al., 2021	Poseidon security analysis	Analysis	—	—
Ben-Sasson et al. (2020) Ben-Sasson et al., 2020	DEEP-FRI	Proof	—	—
Bünz et al. (2018) Bünz et al., 2018	Bulletproofs	Proof	Impl	—
Katz et al. (2018) Katz et al., 2018	MPC-in-the-head NIZK	Proof	Impl	—
Quan (2022) Quan, 2022	Bitcoin PQ + lattice + STARK	—	Impl	—
Argo et al. (2024) Argo et al., 2024	PQ signatures for privacy	Proof	Impl	OMUF+BL
Chathurangi et al. (2025) Chathurangi et al., 2025	Anonymous credentials survey	Survey	—	—
Slamanig (2025) Slamanig, 2025	Theory vs. practice survey	Survey	—	—
Buser et al. (2022) Buser et al., 2022	Exotic PQ sigs survey	Survey	—	—
This project (target)	SPHINCS+–PoS–Fischlin	Proof	Impl+Bench	OMUF+BL